



ITM SKILLS
UNIVERSITY

Restaurant Table Reservation & Waitlist System

Case Study 70 · Java Programming

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Available



Reserved



Occupied

The problem



Double bookings

Two parties promised the same table



Wasted big tables

A couple seated at a six-seater



Walk-ins skipped

Guests forgotten or seated out of turn



GOAL

**Seat by the rules - and
tell the host who is
next**

Objectives and modules

6

objectives

1 Table Setup

5 Availability Notification

7

modules

2 Reservation Booking

6 Reservation Search

3 Waitlist Management

7 Seating Reports

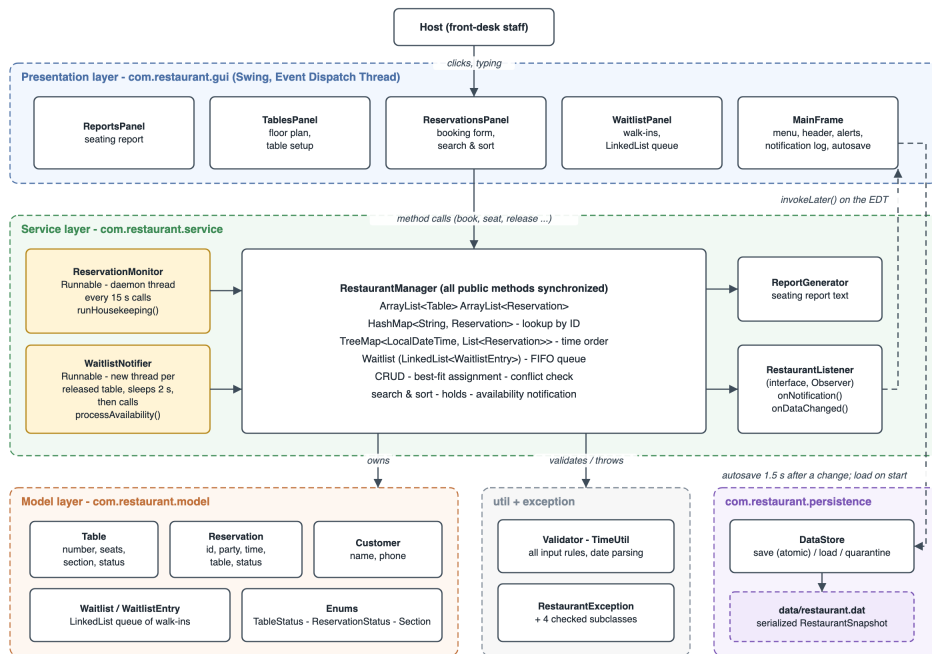
14

Java concepts

4 Table Assignment

Architecture

System Architecture - layers, threads and data flow



GUI → Service → Model

The Swing screens only collect input and show results



One decision-maker

RestaurantManager holds every rule



Two background threads

Notifier (waits 2 s) and monitor (every 15 s)



Saved automatically

data/restaurant.dat, restored on restart

Data structures

ArrayList

All tables and reservations
keeps everything

HashMap

Reservation ID → reservation
O(1) lookup

TreeMap

Bookings sorted by start time
"who is due in 30 min?"

LinkedList

The walk-in waitlist
first come, first served

TableStatus enum

AVAILABLE

RESERVED

OCCUPIED

Key algorithms

1

Best fit

Smallest table that fits the party

2

No double booking

Each party blocks its table for 90 min

3

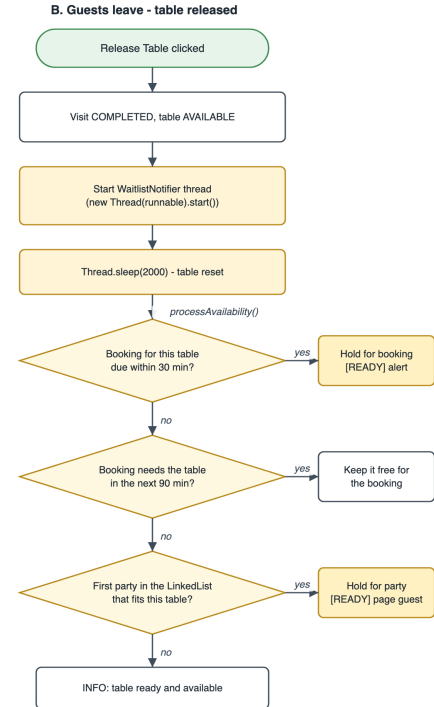
Fair queue

First waiting party that fits gets the table

4

Notification

Release → thread waits 2 s → hold → alert



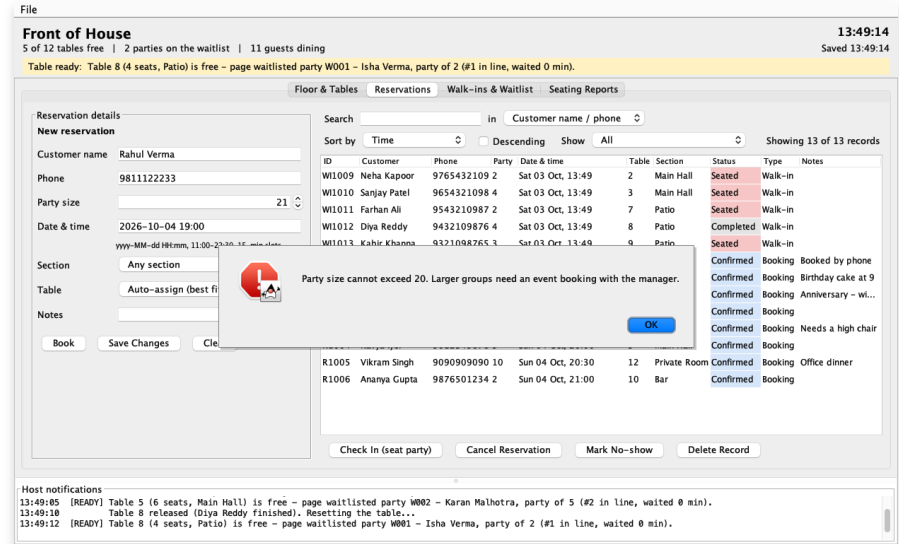
Flowchart 2 - table released

Testing and results

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test cases passed

- ✓ Every module and every validation rule
- ✓ Errors explain the fix: "15-minute slots", "cannot exceed 20"
- ✓ Background notification verified at about 2 s
- ✓ All data restored after a restart



Real screenshot: party of 21 typed into the booking form

Conclusion and future scope

6

objectives met

7

modules built

14

Java concepts used

Plus realistic rules: best fit, a fair queue, held tables and autosave

Next steps



Database

Several front desks share one reservation book



SMS paging

Waiting guests are notified automatically



Join tables

Seat very large groups together

Thank you



Questions?

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